

1.

Let X have an exponential distribution

$$p(x|v) = \begin{cases} ve^{-vx}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Sketch $p(x|v)$ versus x for a fixed value of the parameter v .

(b)

Sketch $p(x|v)$ versus v , $v > 0$ for a fixed value of x .

(c)

Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|v)$. Show that the maximum likelihood estimate for v is given by

$$\hat{v} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}$$

2.

Let X have a uniform distribution

$$p(x|v) = \begin{cases} \frac{1}{v}, & 0 \leq x \leq v \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Sketch $p(x|v)$ versus v for an arbitrary value of x .

(b)

Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|v)$.

Show that the maximum likelihood estimate for v is $\max_k x_k$.

(c)

Find the method of moments estimator for v .

3.

4.

(a)

(b)

5.

6.

(a)

(b)

(c)